

MySQL Storage Engines

Concepts, Verification, Selection, and Practical Applications

DSAI1002 – Database Tutorial

Updated: June 4, 2026

Learning Objectives

After this lesson, learners can

- 1 Explain the concept of a **storage engine** in MySQL.
- 2 Describe the engine's role in storing and processing table data.
- 3 Inspect available engines using `SHOW ENGINES` and `information_schema.engines`.

Practical Skills

- 4 Specify an engine when creating tables using `ENGINE`.
- 5 Differentiate commonly used storage engines.
- 6 Choose the appropriate engine for real-world scenarios.

High-Level Overview

In MySQL

Each table is managed by a specific **storage engine**.

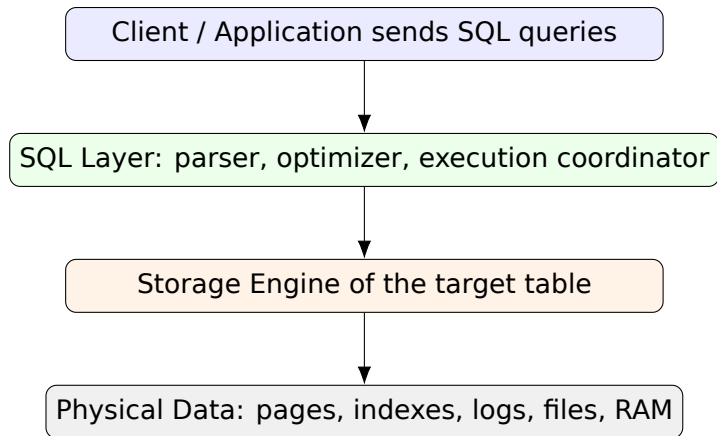
- The engine governs how table data is stored, read, written, locked, and recovered.
- Not all engines support the same set of capabilities.
- For example: InnoDB supports transactions and foreign keys; MEMORY stores data entirely in RAM.

Two Primary Layers

SQL Layer Receives statements, parses syntax, and optimizes query execution plans.

Engine Layer Manages physical storage and data retrieval for each table.

MySQL Server and Storage Engines



Storage engines operate beneath the SQL layer and directly manage table-level data storage.

What is a Storage Engine?

Definition

A **storage engine** is a software module within MySQL responsible for handling data storage, retrieval, and management for individual tables.

Determines Storage Format

- Whether data resides on disk or in system memory.
- How indexing, caching, and crash recovery are organized.

Determines Feature Support

- Whether transactions (ACID) are supported.
- Whether foreign keys and row-level locking are available.

Default Storage Engine

- Since MySQL 5.5, InnoDB has been the default storage engine.
- If a CREATE TABLE statement omits the ENGINE clause, MySQL applies the server default.

```
CREATE TABLE students (  
    student_id INT PRIMARY KEY,  
    student_name VARCHAR(100)  
);
```

Interpretation

If the default engine is InnoDB, the students table will be created using InnoDB automatically.

Specifying the Storage Engine at Creation

General Syntax

```
CREATE TABLE table_name (  
    column_list  
) ENGINE = engine_name;
```

Example: Creating a Table with InnoDB

```
CREATE TABLE customers (  
    customer_id INT PRIMARY KEY,  
    customer_name VARCHAR(100),  
    phone VARCHAR(20)  
) ENGINE = InnoDB;
```

SQL Processing Flow with Storage Engines

- ➊ **Receive SQL statement:** Client issues SELECT, INSERT, UPDATE, DELETE, or CREATE TABLE.
- ➋ **Parse and optimize:** MySQL validates syntax, permissions, and formulates an execution plan.
- ➌ **Call storage engine:** MySQL delegates read/write requests down to the table's designated engine.
- ➍ **Access data:** The engine reads, writes, locks, recovers, or indexes data via its internal mechanisms.
- ➎ **Return results:** MySQL receives the data records from the engine and returns the response to the client.

Checking Available Storage Engines

Method 1: SHOW ENGINES

```
SHOW ENGINES;
```

Method 2: System Table

```
SELECT  
    engine,  
    support  
FROM information_schema.engines  
ORDER BY engine;
```

Key Takeaway

These queries identify which engines are supported, which one is set as default, or if any are disabled.

Support Values in SHOW ENGINES

Value	Meaning
YES	Engine is supported on this server.
NO	Engine is not supported.
DEFAULT	Engine is supported and currently configured as default.
DISABLED	Engine exists in the binary but is currently disabled.

Key Evaluation Criteria

Transactions, foreign keys, crash recovery, persistence, and read/write optimization profiles.

Major Storage Engines in MySQL

Common Engines

- InnoDB: Default engine, transaction-safe (ACID).
- MyISAM: Legacy engine, read-heavy, non-transactional.
- MEMORY: Fast in-memory volatile storage.
- ARCHIVE: Compressed historical archive storage.

Specialized Engines

- CSV: Stores data as plain text CSV files.
- BLACKHOLE: Accepts incoming data without storing it.
- MERGE/MRG_MyISAM: Unions identical MyISAM tables.
- FEDERATED: Connects to remote MySQL servers.

InnoDB: The Default and Standard Engine

Concept

InnoDB is the safest and most capable choice for enterprise systems due to its robust transactional support, referential integrity, and crash resilience.

Transactions

Supports COMMIT, ROLLBACK, and ACID properties.

Foreign Keys

Enforces referential integrity constraints.

Row Locking

Reduces lock contention in concurrent multi-user workloads.

Crash Recovery

Automatic recovery using write-ahead redo logging.

Example: Creating an InnoDB Table

```
CREATE TABLE orders (  
    order_id INT PRIMARY KEY,  
    customer_id INT NOT NULL,  
    order_date DATE,  
    total_amount DECIMAL(12, 2)  
) ENGINE = InnoDB;
```

Recommended Use Cases for InnoDB

- E-commerce, billing, and sales order processing systems.
- Financial accounting, payments, and user identity tables.
- Any table demanding strict data integrity, foreign keys, and error rollbacks.

MyISAM

Characteristics

- Default engine prior to MySQL 5.5.
- Can be fast in specific read-heavy scenarios.
- Does not support ACID transactions.
- Does not support foreign key constraints.

Example

```
CREATE TABLE article_archive (  
    article_id INT PRIMARY KEY,  
    title VARCHAR(255),  
    content TEXT  
) ENGINE = MyISAM;
```

MEMORY

Characteristics

- Stores all table data in RAM.
- Ideal for temporary caching and lookup tables.
- Extremely low-latency reads and writes.
- Volatile: data is wiped when MySQL Server stops or restarts.

Example

```
CREATE TABLE session_cache (  
    session_id VARCHAR(100) PRIMARY KEY,  
    user_id INT,  
    created_at DATETIME  
) ENGINE = MEMORY;
```

ARCHIVE and CSV

ARCHIVE

Stores large volumes of unindexed, append-only historical records in a compressed format; ideal for audit logs.

```
CREATE TABLE audit_logs (  
    log_id INT,  
    action_name VARCHAR(100),  
    action_time DATETIME  
) ENGINE = ARCHIVE;
```

CSV

Stores data directly in text files using comma-separated values; easy for file exchange but lacks indexing and constraints.

```
CREATE TABLE export_customers (  
    customer_id INT NOT NULL,  
    customer_name VARCHAR(100) NOT NULL,  
    phone VARCHAR(20) NOT NULL  
) ENGINE = CSV;
```

BLACKHOLE, MERGE, and FEDERATED

BLACKHOLE

Discards written data while maintaining binary logs. Used in replication relaying and testing.

MERGE / MRG_MyISAM

Combines multiple identically structured MyISAM tables into one logical view.

FEDERATED

Queries tables on remote MySQL instances without holding physical table data locally.

Note

These engines serve specific architectural needs and are not intended for general application tables.

Principles for Choosing a Storage Engine

Data Integrity Requirements

If tables have foreign key relations, require rollbacks, or process financial data, select InnoDB.

Data Durability Needs

If records must persist across server restarts, never use MEMORY for persistent data.

Performance Goals

Temporary calculation tables benefit from MEMORY; robust concurrent read/write workloads belong on InnoDB.

Nature of Data

High-volume write-once audit logs suit ARCHIVE; external exports may use CSV.

Practical Use Cases

Scenario	Suitable Engine
Sales & E-commerce	InnoDB: Demands transactions, rollback, foreign keys, and ACID guarantees.
Temporary Caching	MEMORY: Fast short-lived intermediate data storage.
Log Archiving	ARCHIVE: High-volume historical records, minimal updates, compact storage.
Data Interchange	CSV: Easily read by spreadsheet applications and external export scripts.
Special Testing	BLACKHOLE: Captures write commands without consuming disk capacity.

Role in Database Administration

Database Design

Select engines tailored to each table's role; primary business tables should consistently use InnoDB.

Performance Tuning

Understanding engine mechanics assists in diagnosing slow queries, lock contention, and I/O bottlenecks.

Backup and Recovery

Different engines use distinct on-disk formats, requiring customized backup and restore workflows.

System Operations

During version upgrades or server migrations, DBAs must verify supported and default engines.

Quick Comparison: Common Engines

Criteria	InnoDB	MyISAM	MEMORY	ARCHIVE
Transactions	Yes	No	No	No
Foreign Keys	Yes	No	No	No
Data Durability	Yes	Yes	No	Yes
Primary Business Data	Yes	Limited	No	Limited
Primary Use Case	OLTP / Business	Read-heavy / legacy	Staging tables	Logs / history

Practical Guideline

For the vast majority of modern applications, InnoDB is the safest and recommended default.

Quick Comparison: Specialized Engines

Criteria	CSV	BLACKHOLE	FEDERATED
Transactions	No	No	No
Foreign Keys	No	No	No
Data Durability	Yes	No	Depends on remote server
Primary Business Data	No	No	Limited
Primary Use Case	CSV file export	Discard writes / replication	Remote table access

Caution

Specialized engines should only be deployed when there is an explicit architectural justification.

Quick Quiz 1

Question 1

To which entity is a MySQL storage engine directly attached?

- A. User account
- B. Table
- C. Database password
- D. SQL comment

Quick Quiz 2

Question 2

Why is understanding storage engines essential?

- A. Because the engine defines the entire SQL syntax.
- B. Because each engine has distinct storage characteristics and capabilities.
- C. Because MySQL supports only a single storage engine.
- D. Because engines are only used for renaming tables.

Quick Quiz 3

Question 3

Which storage engine is default in modern MySQL releases?

- A. MyISAM
- B. MEMORY
- C. InnoDB
- D. CSV

Quick Quiz 4

Question 4

Which clause is used to specify the storage engine when creating a table?

- A. USING
- B. ENGINE
- C. STORAGE PATH
- D. TABLE TYPE ONLY

Quick Quiz 5

Question 5

Which command lists all available storage engines on the server?

- A. SHOW TABLES;
- B. SHOW ENGINES;
- C. SHOW DATABASES;
- D. SHOW USERS;

Quick Quiz 6

Question 6

What does the value DEFAULT signify in SHOW ENGINES?

- A. The engine is disabled.
- B. The engine does not exist.
- C. The engine is currently configured as the server default.
- D. The engine is used exclusively for backups.

Quick Quiz 7

Question 7

Which type of system is InnoDB best suited for?

- A. Systems requiring transactions, rollbacks, and foreign keys.
- B. Tables used only to discard write data.
- C. CSV files opened directly in Excel.
- D. Ephemeral tables that lose data upon server shutdown.

Quick Quiz 8

Question 8

Which storage engine stores all its data directly in RAM?

- A. MEMORY
- B. CSV
- C. BLACKHOLE
- D. FEDERATED

Quick Quiz 9

Question 9

Which engine receives data without persisting it to disk or memory?

- A. InnoDB
- B. BLACKHOLE
- C. MyISAM
- D. ARCHIVE

Quick Quiz 10

Question 10

If a table requires foreign key integrity, which engine is most suitable?

- A. InnoDB
- B. MEMORY
- C. BLACKHOLE
- D. CSV

Quick Quiz 11

Question 11

Which engine is well-suited for large, append-only, infrequently updated logs?

- A. ARCHIVE
- B. BLACKHOLE
- C. FEDERATED
- D. MEMORY

Review Questions

- 1 What purpose does a storage engine serve in MySQL?
- 2 Why is InnoDB generally recommended for core business entities?
- 3 Differentiate InnoDB and MEMORY regarding data persistence.
- 4 Provide a realistic scenario where ARCHIVE is an optimal choice.
- 5 Why should BLACKHOLE never be chosen for data requiring storage?

Practical Exercises 1-2

Exercise 1

Write an SQL statement to list all storage engines available on your MySQL Server.

Exercise 2

Create a table named `students` with columns `student_id`, `student_name`, and `email`. The table must explicitly use InnoDB.

Requirement

Use standard SQL syntax and explicitly specify the engine when creating the table.

Practical Exercises 3-4

Exercise 3

Create a table named `temp_report` with columns `report_id`, `report_name`, and `created_at` using `MEMORY`. Explain its major operational limitation.

Exercise 4

An e-commerce system needs to store customers, orders, and payments. It requires transaction rollbacks on errors and strict foreign key integrity.

Requirement

Select the most suitable storage engine and justify your technical rationale.

Lesson Summary

- Storage engines determine how MySQL stores, formats, and retrieves data per table.
- InnoDB is the default engine and standard choice for modern business applications.
- InnoDB provides ACID transactions, foreign keys, row locking, and automatic crash recovery.
- MEMORY delivers high-speed in-RAM access but is volatile across server restarts.
- ARCHIVE is specialized for compressed, append-only historical log datasets.
- CSV, BLACKHOLE, MERGE, and FEDERATED address specialized technical needs.

Key Terms

- MySQL
- Storage engine
- InnoDB
- MyISAM
- MEMORY
- ARCHIVE
- CSV
- BLACKHOLE
- MERGE
- FEDERATED
- Transaction
- Foreign key
- SHOW ENGINES
- CREATE TABLE
- ENGINE

Quick Quiz Answer Key

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11
B	B	C	B	B	C	A	A	B	A	A

Self-Check Guidance

For any incorrect answer, review the corresponding slide to clarify the underlying concept: default engines, ENGINE syntax, durability, transactions, or foreign keys.

Exercise 1 Solution

Method 1

```
SHOW ENGINES;
```

Method 2

```
SELECT  
    engine,  
    support  
FROM information_schema.engines  
ORDER BY engine;
```

Exercise 2 Solution

```
CREATE TABLE students (  
    student_id INT PRIMARY KEY,  
    student_name VARCHAR(100),  
    email VARCHAR(100)  
) ENGINE = InnoDB;
```

Best Practice

For production tables, explicitly declaring `ENGINE = InnoDB` avoids unexpected behavior if server default configurations change.

Exercise 3 Solution

```
CREATE TABLE temp_report (  
    report_id INT PRIMARY KEY,  
    report_name VARCHAR(100),  
    created_at DATETIME  
) ENGINE = MEMORY;
```

Limitation

Data in MEMORY tables is volatile and will be lost whenever MySQL Server restarts or shuts down.

Exercise 4 Solution

Recommended Storage Engine

Select InnoDB.

Rationale

- The workload requires ACID transactions with rollbacks upon execution errors.
- Orders must reference valid customers through foreign key constraints.
- Customer, order, and payment data require strict relational consistency.
- InnoDB provides crash recovery through redo logging for mission-critical business data.